

DSAIN

Distributed Sovereign AI Infrastructure Nodes Technology

Concept Development Document

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Classification: Concept Development Document

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Executive Summary

DSAIN (Distributed Sovereign AI Infrastructure Nodes Technology) is a proposed European infrastructure initiative focused on the development of a distributed network of sovereign AI infrastructure nodes.

The initiative addresses increasing demand for artificial intelligence, high-performance computing, cloud services, research computing, and digital sovereignty by creating a resilient infrastructure architecture that avoids excessive concentration of computing resources in a limited number of hyperscale locations.

The DSAIN model is based on infrastructure ownership rather than ownership of rapidly depreciating AI hardware. The project seeks to establish regional infrastructure nodes capable of hosting AI systems, cloud services, research workloads, and future digital applications while supporting regional economic development and technological resilience.

The first implementation is envisioned as Demonstrator Node 01 in Bedburg-Hau, Germany.

Vision

To create a resilient European network of distributed sovereign AI infrastructure nodes supporting the next generation of artificial intelligence, cloud computing, research, and digital services.

DSAIN aims to strengthen:

- European technological sovereignty
- Infrastructure resilience
- Regional innovation ecosystems
- Sustainable digital infrastructure

- Research and industrial competitiveness
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Mission

Develop, own, and operate a scalable network of AI infrastructure facilities that provide secure, efficient, and sustainable environments for advanced computing workloads across Europe.

Strategic Rationale

Artificial intelligence is becoming a critical economic and strategic capability.

Current computing infrastructure is increasingly concentrated in a limited number of hyperscale locations, creating challenges related to:

- Grid dependency
- Geographic concentration
- Infrastructure resilience
- Digital sovereignty
- Regional inequality

DSAINTE proposes an alternative model based on distributed regional infrastructure nodes.

Infrastructure-as-a-Platform Model

DSAINTE is not primarily an AI model company.

Instead, DSAINTE operates as an Infrastructure-as-a-Platform (IaaS) organization.

DSAINTE owns and operates:

- Land
- Buildings
- Electrical infrastructure
- Cooling systems
- Security infrastructure
- Connectivity infrastructure
- Facility operations

Customers provide:

- GPUs
- AI accelerators
- Cloud platforms
- AI applications
- Research workloads

Demonstrator Node 01

Location

Bedburg-Hau, North Rhine-Westphalia, Germany

Planned Capacity

Initial Capacity:

2 MW IT Load

Expansion Capacity:

Up to 6 MW

Technical Classification

Tier III Infrastructure

Target Availability:

99.982%

Redundancy:

N+1

Target PUE:

≤ 1.20

Site Components

Demonstrator Node 01 includes:

1. AI Infrastructure Core
2. Admin & Operations Building
3. Maintenance & Storage Building
4. Dedicated 20kV / 110kV Substation
5. Technical Cooling Plant
6. Parking Area
7. Security Airlock System
8. Main Entrance
9. Service Entrance

10. Employee Housing Building
 11. Children's Playground
 12. Guest Center
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Technical Architecture

Computing Infrastructure

Support for:

- AI Training
- AI Inference
- HPC Workloads
- Research Computing
- Cloud Services

Cooling

Direct-to-Chip Liquid Cooling

Benefits:

- Higher rack density
- Improved efficiency
- Reduced power consumption

Network Architecture

- Multi-carrier connectivity
 - Redundant fiber infrastructure
 - High-bandwidth backbone
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Sustainability Strategy

DSAINTE integrates sustainability from the beginning.

Geothermal BTES System

Borehole Thermal Energy Storage:

- 12–20 boreholes
- 120–180 meters depth
- Thermal balancing
- Seasonal energy storage

Heat Reuse

Potential future applications:

- District heating
 - Agricultural heating
 - Public infrastructure support
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Security Architecture

The infrastructure is designed around a dual-perimeter security model.

Physical Security

- Controlled access
- CCTV monitoring
- Airlock entry systems
- Security Operations Center

Digital Security

- Network segmentation
 - Continuous monitoring
 - Secure connectivity architecture
 - Infrastructure resilience measures
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Business Model

Revenue may be generated through:

- Rack leasing
 - Infrastructure hosting
 - Cloud infrastructure services
 - Research partnerships
 - Enterprise hosting
 - Connectivity services
 - Energy partnerships
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Governance Model

DSAINTE is envisioned with a structured governance framework.

Supervisory Board

Strategic oversight and accountability.

Executive Team

- Chief Executive Officer
- Chief Operations Officer
- Chief Technology Officer
- Chief Financial Officer
- Chief Security Officer

Technical Governance Board

Infrastructure standards and architecture.

Security Committee

Security policy and risk management.

Sustainability Committee

Environmental and energy performance oversight.

Expansion Roadmap

Phase 1

2026–2030

Demonstrator Node 01

Bedburg-Hau, Germany

Phase 2

2030–2033

Expansion to 3 German regional nodes

Phase 3

2033–2036

Expansion to 6 German nodes

Phase 4

2036–2040

Development of a broader European infrastructure network

Target:

10+ interconnected infrastructure nodes

Economic Impact

Potential benefits include:

- Regional job creation
 - Infrastructure investment
 - Research collaboration
 - SME ecosystem development
 - Technology sector growth
 - Educational partnerships
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Potential Partnerships

DSAINTE seeks engagement with:

- Municipalities
 - Universities
 - Research institutions
 - Engineering firms
 - Utilities
 - Technology companies
 - Infrastructure investors
 - European innovation programs
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Long-Term Vision

By 2040, DSAINTE aims to operate a distributed network of sovereign AI infrastructure nodes across Europe, supporting advanced computing workloads while strengthening digital sovereignty, resilience, sustainability, and regional economic development.

Contact

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DSAINTE

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Concept Development Initiative

Building the Infrastructure of Europe's AI Future.